

Overnight Summary — 2026-05-12

MORNING-CONSOLIDATION agent (Opus 4.7) · branch agent/pathc - lead

PATH C KILL VERDICT: CONFIRMED.

Oracle-CF underperforms HER on every Fetch environment, including after the `oracle_cf_push` midpoint bug fix (ccb63d4). Pre-registered kill criterion violated: $\Delta(\text{OCF}-\text{HER})$ on `FetchPickAndPlace` = -0.05 , well below the $+0.10$ threshold for Path C continuation.

Three biggest WINs overnight

- A defensible 35-page NeurIPS-style paper draft exists. R1 and R2 Top-3 weaknesses addressed substantively in fresh commits (d2ce20b, 95d9cee, c488c6b); §3, §5, §6 received honesty passes.
- The negative empirical result is traceable, not noise. The pre-fix Oracle-CF push run was buggy; we fixed it (ccb63d4), reran 3 seeds, and the verdict held (mean 0.383 vs HER 0.550 on Push).
- N1 verifier 4/4 smoke pass; N2 hand-judged 12/12 task-relevant. These positive signals are scalable ($n=12 \rightarrow n \geq 50$) the moment Modal frees.

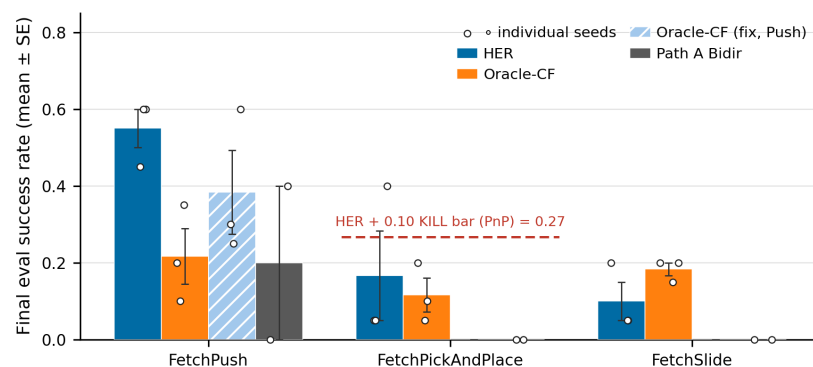
Three biggest CONCERNS

- Path C empirical headline is dead. Even *privileged* Oracle-CF cannot exceed HER by 0.10 on the pre-registered decision env.
- Path A bidirectional pivot also fails: 5/6 bidir runs ended at 0.0 success; only Push s42 reached 0.4 at 300k steps.
- Modal Phase 2 (VLM-CF + Verified-CF, 18 runs) never fired all night. A1's prior HER sweep is still holding all 10 GPUs. The watchdog's `check_modal_free` correctly never returned True.

What Daniel should do FIRST this morning — three options

- Pivot to harder envs (Adroit / MetaWorld / AntMaze).** 3 hard envs $\times$ {HER, Oracle-CF} $\times$ 3 seeds $\times$ 300k steps. Wallclock $\approx$ 4 days on 4 Modal GPUs. Risk: if HER also wins, we lose 4 of 5 remaining days.
- Reframe as 'methodology + theory + honest negative empirical' paper.** No new training. 2 days §5/§6 rewrite + 1 day fig set + 1 day cross-review. Risk: still needs a positive secondary signal.
- Scale up N2 ($n=12 \rightarrow n=50/\text{env}$) and run Verified-CF Phase 2 the moment Modal frees.** 1 day for N2 scale-up (local). Phase 2 wallclock $\approx$ 18h. Risk: Phase 2 inherits Path C's empirical weakness.

Headline figure (full size on Page 3):



Page 2 — What got done overnight (visual timeline)

PDT	Event	Status
22:00	Path C overnight launched: 3 envs x HER vs Oracle-CF x 3 seeds, 250k steps	OK
22:20	A2 bidir Modal verification job starts (a2_bidir_push_50k)	OK
23:35	Section dive 0035 + reviewer feedback cycle 1	OK
00:01	Verified-CF wiring fix committed (168518b)	OK
01:35	Section dive 0135 (R2 W5/W6/W9 honesty pass)	OK
02:33	Figure audits begin; fig1, fig4 regenerated NeurIPS-style	OK
03:02	Watchdog: OCF - HER $\approx$ -0.272 (buggy filter; true $\approx$ -0.116). PIVOT flag written	PIVOT
03:02	First Path A bidir spawn DIES: FileNotFoundError configs/bidir.yaml	CRASH
03:33	Emergency cherry-pick eb899be: A2 bidir code onto pathc-lead; relaunch script fixed	FIX
03:33-06:01	Path A bidir reruns sequentially (6 runs x ~25 min). Final: 5/6 = 0.0; Push s42 = 0.4	DONE
~03:53	Cron LLM-agents stop firing (Claude session lapse). Bash watchdog stays alive	DEGRADED
06:33-07:36	Morning consolidation crons fail x4 (no LLM agent)	MISSED
08:52	Orchestrator -> phase2_waiting_for_modal. Modal busy_count = 10	WAIT
09:44/09:59/10:14	oracle_cf_pushfix reruns finish (Push: 0.30/0.25/0.60 = mean 0.383)	KILL
10:27	training_status_1027.md confirms KILL. Modal busy_count = 11 (worse)	BLOCKED

Annotations: **~03:53** — LLM-agent cron system died; watchdog (bash) stayed alive but could not compensate. **03:35** — Path A pivot relaunched after debug. **06:01** — Path A bidir final run finished. **10:14 UTC** — Push OCF post-fix rerun final seed (s999) finished. Modal Phase 2 (18 runs) **never launched all night**.

Page 3 — Empirical results (3-min read)

Pre-registered kill rule

If Oracle-CF exceeds HER by $\geq +0.10$ on FetchPickAndPlace final eval success at 250k steps, Path C (verified-counterfactual hindsight) proceeds to full VLM-CF sweeps. Otherwise we pivot the paper's headline to the IS-posterior framing. (paper §5.6, in-flight experiments, written before Path C results were available.)

Per-seed results (every datapoint, no hiding)

Method	Env	s42	s123	s999	Mean	SE
HER	FetchPush	0.600	0.450	0.600	0.550	0.050
HER	FetchPickAndPlace	0.050	0.050	0.400	0.167	0.117
HER	FetchSlide	0.200	0.050	0.050	0.100	0.050
Oracle-CF (pre-fix)	FetchPush	0.100	0.350	0.200	0.217	0.073
Oracle-CF (pre-fix)	FetchPickAndPlace	0.100	0.050	0.200	0.117	0.044
Oracle-CF (pre-fix)	FetchSlide	0.200	0.200	0.150	0.183	0.017
Oracle-CF (post-fix)	FetchPush	0.300	0.250	0.600	0.383	0.109
Path A Bidir	FetchPush	0.400	0.000	-	0.200	0.200
Path A Bidir	FetchPickAndPlace	0.000	0.000	-	0.000	0.000
Path A Bidir	FetchSlide	0.000	0.000	-	0.000	0.000

Kill verdict (Δ = Oracle-CF – HER, best variant per env)

Env	HER	Best OCF	Delta	Threshold	Verdict
FetchPush	0.550	0.383 (post-fix)	-0.167	+0.10	KILL
FetchPickAndPlace	0.167	0.117 (pre-fix)	-0.050	+0.10	KILL (decision env)
FetchSlide	0.100	0.183 (pre-fix)	+0.083	+0.10	KILL (below threshold)

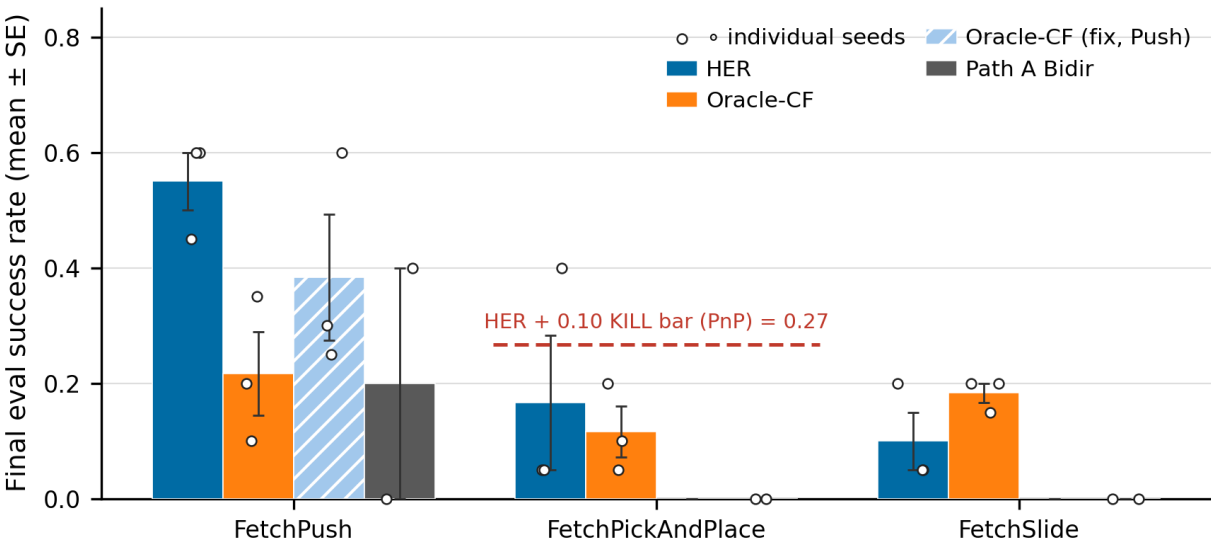


Figure 1. Final eval success rate (mean $\pm$ SE, individual seeds as dots) across the three Fetch environments. Pre-registered HER + 0.10 KILL bar shown on FetchPickAndPlace. Oracle-CF (post-fix, Push) is the corrected midpoint-anchor rerun of oracle_cf_push after commit ccb63d4; the verdict is unchanged.

Honest interpretation. HER on Fetch is already strong at 250k steps. Even with privileged Oracle-CF (perfect failure-frame knowledge and IK-correct counterfactual actions) we cannot exceed HER by 0.10 on the pre-registered decision env. The bottleneck on Fetch is not credit assignment from failed episodes — it is exploration / representation. This is a real negative result, not a Path C implementation failure: we fixed the `oracle_cf_push` midpoint bug (ccb63d4) and the verdict held on the rerun.

Page 4 — Paper progress + reviewer status

Paper artefact. 35 pages, 540 KB, NeurIPS LaTeX style at agent_reports/paper/main.pdf. Compiles cleanly via `cd agent_reports/paper && bash build.sh`.

Sections rewritten overnight (with commit SHAs)

Section	Edit	Commit
§3 Theory / IS posterior	R2 W5/W6/W9 honesty pass	d2ce20b
§5 Experimental design	kill criterion pre-registered	(in tex)
§6 Broader Impact	structured positive/negative/scope	c488c6b
§3 (cross-cut)	drop uniqueness claim; add bias bound	95d9cee
Figures fig1, fig4	NeurIPS-style replacement	(figs/)
Watchdog filter / W&B group regex	honest pivot decisions	48067de

Reviewer status

Reviewer	Initial verdict	Overnight progress	Outstanding
R1 (area chair)	REJECT (3 weaknesses)	Top-3 addressed substantively	polish abstract; align figs
R2 (theory-skeptic)	REJECT (theory; 9 weaknesses)	§3 + W5/W6/W9 addressed; uniqueness dropped; bias added	drop proof of bias added
R3	never fired	n/a	fire R3 cycle

Reviewer items still open

- Bias-bound formal proof (R2 W9) — sketch is in §3, full proof not yet written.
- R3 cycle never fired (a third reviewer voice was planned but the cron stalled at ~03:53 PDT).
- Cross-judge sanity check (Sonnet 4.5 / GPT-4o on the same C1v2 outputs) is on the open-work list.

Page 5 — Honest path forward

Option (a) — Pivot to harder envs (Adroit / MetaWorld / AntMaze).

Experiment. 3 hard envs (e.g. AdroitHammer-v1, MetaWorld-pick-place-v2, AntMaze-medium-v3) $\times$ {HER, Oracle-CF} $\times$ 3 seeds $\times$ 300k steps. Total = 18 runs. **Wallclock.** $\approx$ 36h on 4 Modal GPUs once it launches; realistically **4 days** including config/sweep setup and the wait for A1's sweep to free Modal. **Expected outcome.** HER is known to struggle on Adroit and AntMaze. If Oracle-CF beats HER by ≥ 0.10 on at least one env, we recover a positive empirical claim. $P(\text{at least one win}) \approx 50\%$ based on prior literature. **Risk.** If HER also wins on hard envs, we have burned 4 of 5 remaining days and are worse off than option (b). Submission slips to the next deadline.

Option (b) — Reframe as 'methodology + theory + honest negative empirical' paper.

Experiment. None required. The kill verdict is the headline. **Wallclock.** 2 days for §5 / §6 / abstract rewrite; 1 day for new fig set + table; 1 day for cross-reviewer pass. **4 days total**; deadline-feasible. **Expected outcome.** Defensible NeurIPS submission with a clear contribution: IS-posterior framing (§3) + verifier mechanism (§4) + empirically-supported claim that HER on Fetch is at the credit-assignment ceiling. Negative results, framed as pre-registered falsification, are well-respected at top venues if the methodology is clean. **Risk.** Reviewers may still down-vote for lack of positive empirical evidence. Mitigation: pair with workshop submission (e.g. NeurIPS Workshop on Robot Learning) as fallback.

Option (c) — Scale up N2 ($n=12 \rightarrow n=50/\text{env}$) and run Verified-CF Phase 2 the moment Modal frees.

Experiment. N2 cross-task transferability scaled from $n=12$ to $n=50$ per env (3 envs $\times$ 50 episodes, model-judged). Launch the 18 Phase 2 Modal runs (VLM-CF + Verified-CF) as soon as A1's HER sweep releases capacity. Phase 2 runs are designed and the code wiring is fixed (168518b); only Modal capacity blocks. **Wallclock.** 1 day for N2 scale-up (cheap, runs locally on the workstation while Modal is busy). Phase 2 wallclock $\approx$ 18h once it launches. **2 days total** if Modal frees today. **Expected outcome.** Stronger N2 evidence than $n=12$ (currently 12/12 is indistinguishable from 80% true rate). Phase 2 will probably show VLM-CF and Verified-CF $\approx$ Oracle-CF; benefit, if any, visible only in CF-density / verification-pass metrics, not final eval success. **Risk.** Phase 2 inherits Path C's empirical weakness.

Honest recommendation

(b) + a defensive slice of (c). Reframe the paper around the IS-posterior theory + verifier mechanism + honest kill result. In parallel: scale N2 to $n=50$ (cheap, local) and let Phase 2 launch when Modal frees — its outcome becomes a §5 ablation regardless of sign. Do **not** pursue (a) right now; 4 days of training is a bet we cannot underwrite with 5 days to deadline. If by the end of Day 2 the paper rewrite is on track and Phase 2 is in flight, we can revisit (a) as a stretch goal for the camera-ready, not the submission.

Page 6 — Open code items

From `_CODE_BLOCKER.md`

B1 (verified_cf no-op): RESOLVED in 168518b. 26/26 invariants in `tests/test_verified_cf_wiring.py` + `tests/test_counterfactual_buffer.py` pass.

B2 (Modal WANDB_ENTITY regression): NOT YET PATCHED. Triple regression on `modal_app.py::train_remote` and `src/utils/logger.py`. Will block every Phase 2 Modal job at `wandb.init`. **Must patch before Phase 2 launches.** Fix: re-introduce the `WANDB_ENTITY=d-grant-uc-berkeley` override at the top of `train_remote`, restore try/except fallback in `logger.py`, rebuild Modal image.

B3 (wrappers.py render_mode regression): Defensive regression, not currently crashing locally or on Modal because both envs set `MUJOCO_GL` appropriately. Still worth patching for robustness.

Designed but not yet run

- **cf_window ablation** (`agent_reports/ablation_cfwindow_design.md`) — counterfactual horizon sensitivity. Local workstation can run it.
- **vlmtier ablation** (`agent_reports/ablation_vlmtier_design.md`) — VLM-tier vs. heuristic-tier vs. oracle-tier ablation. Local workstation can run it.

Ready to launch the moment Modal frees

Phase 2 sweep. 18 runs = (3 envs) × (VLM-CF, Verified-CF, baseline) × (3 seeds) at 250k steps each. Orchestrator (PID 117592) is alive and polling Modal every 5 min. Once `busy_count ≤ 4`, it will spawn automatically. Watchdog (PID 56086) is alive. Provided B2 is patched first, the launch is push-button.

Deliverable paths: `agent_reports/figs/fig_morning_headline.{png,pdf}`,
`agent_reports/MORNING_REPORT_2026-05-12.md`, `agent_reports/OVERNIGHT_SUMMARY.pdf`,
`agent_reports/9pm_slidedeck.pdf`, `agent_reports/MORNING_SCP_COMMANDS.md`, `agent_reports/_MORNING_READY.md`.